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# Beyond 200-Gb/s PAM4 ADC and DAC-Based Transceiver for Wireline and Linear Optics Applications

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**ABSTRACT** System considerations, circuit architecture, and design implementation of wireline and linear optics transceivers capable of supporting data-rates beyond 200 Gb/s are presented. We showcase the silicon results of a transceiver designed in the advanced 3-nm CMOS process, which supports long-reach channels with up to 40 dB of loss at Nyquist. These results demonstrate the technology's benefits of doubling the data rate of transceivers while achieving efficiency gains in power consumption and silicon area. This article highlights several key circuits architecture, such as hybrid continuous-time linear equalizer, inductive peaking clock routing, and one stage TX driver based on grounded switches. The proof-of-concept demonstration of 224 Gb/s with linear optics opens the avenue for power-efficient, low-latency future optical communication. This is crucial for high-performance computing (HPC) networking as well as emerging applications in high-end FPGA.

**INDEX TERMS** 224 Gb/s, 3 nm, digital DFE PLL, digital feed-forward equalizer (FFE), direct drive linear optics, electro-optics, PAM4, pulse amplitude modulation (PAM), SAR, SerDes, time-interleaved ADC, transmitter (TX).

## I. INTRODUCTION

SINCE the adoption of ADC-based SerDes PHYs, the technology has benefited from process scaling to double the throughput of such PHYs every 2 to 3 years. The recent growth in artificial intelligence and machine learning technologies and their datacenter demands have necessitated rapid throughput growth.

However, as speeds of such PAM4 PHYs exceeded 200 Gb/s with sampling frequencies exceeding 100 GS/s, the technology had to come up with new and innovative solutions to overcome the signal integrity limitations, to improve the signal-to-noise ratios (SNRs) of the ADC in the receiver (RX) and the DAC in the transmitter (TX) and to employ advanced digital signal processing (DSP) techniques to better equalize challenging wireline channels.

These advancements have given way to a few generations of 224-Gb/s PAM4 SerDes PHYs, which are not only supporting wireline transceiver channels with over 40-dB loss at Nyquist frequency, but also have been shown to support linear optic solutions supporting kilometers of optical cable lengths [1].

The ADC-based receiver (Fig. 1) enables using powerful digital symbol and sequence detection techniques in the DSP backend [2]. This structure follows the 112-Gb/s DSP-based transceivers trend but with architecture and design optimization for 2× higher bandwidth requirements. The

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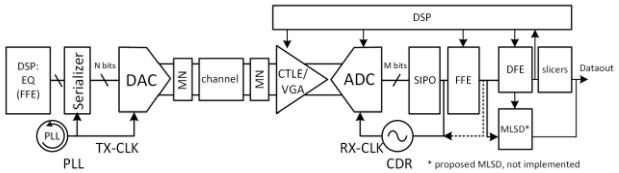



**FIGURE 1.** ADC-based wireline transceiver block diagram.

TX consists of a 9-tap digital feed-forward equalizer (FFE), data serializer, 7-bit DAC with pseudo-differential current mode logic (CML) driver, LC-PLL, clock distribution, and output matching network [3]. The RX includes the matching network, front-end continuous-time linear equalizer (CTLE) and variable gain amplifier (VGA), time-interleaved SAR-ADC, clock distribution with phase interpolator (PI), DCO-based clock data recovery (CDR), and digital equalizers. The digital equalizers consist of an up-to 30-tap digital FFE and optional digital DFE. The FFE is adapted by the sign-sign least mean square algorithm and managed through an on-die micro-controller. The CDR input can be driven either directly from the ADC output or by the output of the FFE [4].

Requirements for the digital equalizer and performance of the analog components were specified by system simulations and IEEE's channel operating margin (COM) compliance. Advanced DSP techniques, such as the maximum likelihood sequence detection (MLSD) in addition to the digital FFE and DFE, have been explored in this context.

Direct drive linear optics enhance current SerDes technology by replacing part of the electrical link with an optical link [1]. This technology results in lower power consumption and decreased latency compared to traditional electro-optical modules, while also extending channel reach from 1 m to over 1 km.

This article is arranged as follows. Section II explores the system considerations and advanced DSP techniques. Sections III and IV describe the overall circuit designs of 224-Gb/s RX (Section III) and TX (Section IV) which were fabricated in a 3-nm process. Section V explores the signal integrity challenges for wireline transceivers beyond 200 Gb/s on realistic end-to-end channels while Section VI presents an alternative direct drive linear optics SerDes for optical channels which are much longer than the classical wireline channels. Section VII presents the measurement results of the fabricated transceiver followed by a conclusion of this article.

## II. SYSTEM CONSIDERATIONS

Traditionally, symbol-by-symbol decoding is employed to map the sampled signal back to the nearest constellation point. This was done both for analog-based SerDes, where the sampling point of the analog value is also the symbol decision point and for an ADC-based SerDes where a digitally represented analog value is processed through digital slicers. The symbol-by-symbol decoding is preferred due to its simplicity and the ability to perform parallel decoding for a vector of symbols. In fact, the main goal of the equalization process is to minimize the inter symbol interference (ISI), resulting in optimized sole symbol detection. In addition, the relation between the achieved SNR and bit error rate (BER) is well defined and can be pre-evaluated by industry standardized flows, such as the COM methodology in IEEE 802.3 standards [5].

As the demand for higher data rates constantly increases while signal integrity and channel improvement are slow to catch up, the achievable SNR with the current SerDes architecture does not meet the target BER as defined by industry standards (IEEE, OIF, etc.). To address this issue, stronger forward error correcting codes (FEC) can be used to relax the pre-FEC BER requirement. This comes with the substantial overhead of complexity resulting in increased power consumption and latency of the link.

If SNR is not sufficient to meet the link's BER requirements with return to symbol-by-symbol detection utilizing FFE only, a DFE equalizer in additional to FFE is used. Another approach for tackling insufficient SNR is a paradigm shift from symbol-by-symbol to sequence decoding, such as the MLSD (Fig. 1). Applying sequence decoding for channels with ISI was first introduced more than 50 years ago by Forney [6], yet, adopted to and implemented in high-speed SerDes systems only recently.

The core idea of the MLSD is to refer to transmitted data symbols as sequence, as opposed to single symbols that are transmitted serially, where the correlation between adjacent data symbols is caused by the channel ISI, instead of a specific code, as in traditional MLSD implementation. In other words, instead of removing ISI, the decoder is utilizing the ISI in the decoding process, for example, an effective $1 + \alpha D$ channel is formed where the ISI level of the first post-cursor is $\alpha$. The Viterbi decoder is used as a feasible method for the MLSD. Although a theoretical 3 dB increase of the SNR over symbol-by-symbol detection is expected for $\alpha = 1$, the actual SNR benefit of the MLSD in a realistic ADC-based SerDes is much less, typically lower than 2 dB. This is caused by several factors, such as the ADC's dynamic range limitation, due to the RX's nonlinearity and the ADC's effective number of bits (ENOBs) as well as noise correlation which decreases the effective value of allowable $\alpha$. For the MLSD to process ISI, the ADC needs to sample it correctly first, so it requires a higher dynamic range than when linear equalization deals with the pre- and post-cursors prior to the ADC. Furthermore, the maximal benefit achieved in a root-raised-cosine (RRC) or Nyquist pulse's partial response is not achievable on ISI channels where FFE is utilized in tandem to MLSD to cancel the pulse tail and as a side effect—enhances the noise [7]. In addition, the Viterbi decoder implementation constraints result in suboptimal sequence and traceback lengths due to lack of truncation symbols, which is not part of the Ethernet standard.

A simulation based on the transceiver's system model was conducted in which the transceiver was tested against

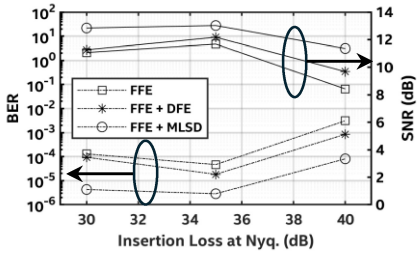



**FIGURE 2.** BER/SNR versus channel's insertion loss at Nyquist for several equalization schemes.

three channel configurations; a reflective channel based on a measured transfer function with a 30-dB insertion loss at Nyquist, and two synthesized nonreflective channels with 35 and 40-dB insertion losses at Nyquist. Under these three scenarios, the system was simulated with three equalization configurations: 1) FFE only; 2) FFE with DFE; and 3) FFE with MLSD. The results shown in Fig. 2 indicate that while reflective real channels provide worse results even at lower insertion loss than synthesized smooth channels due to the additional ISI which the CTLE cannot equalize, the added ISI allows the MLSD to provide higher BER and SNR improvement of about 2 dB. Similarly, when comparing synthesized smooth channels at different insertion losses, channels with higher insertion loss end up with higher residual ISI after FFE resulting in a more effective MLSD.

The effective contribution of the MLSD is still being investigated and debated in the industry regarding its correct integration into the IEEE standard and its effect on the COM calculations. When considering the MLSD gain in the IEEE COM methodology [8], several additional effects may be considered. For example, it is worthwhile to consider noise coloring and enhancement by the tandem FFE/DFE [7], ADC ENOB and maximal dynamic range allowance and RX nonlinearity that effects the PAM4 constellation in outer eyes. Refraining to do so will reflect on MLSD SNR gain that may not be realizable in real receivers, and thus, the COM methodology may qualify channels based on too optimistic expectations.

## III. 224G RECEIVER

The overall RX presented in Fig. 3 consists of an analog front-end (AFE) based on a hybrid $Q$-shaping CTLE followed by a 64-way 6/7-bit resolution time-interleaved ADC and a digital equalizer [4]. A Muller–Muller-based digital CDR is used to recover the clock and control the LC-DCO.

To balance speed, complexity, power, and performance, a 64-way interleaving was used with 16-way interleaved sampling switches in the first stage (TH1) followed by 4-way interleaved sampling switches per each TH1 in the second stage (TH2). As shown in Fig. 3, this architecture uses sixteen 7-GHz clocks with one unit-interval (UI) separation and a 25% duty-cycle for TH1, and 64 1.75-GHz clocks with a 12.5% duty-cycle for TH2.

The AFE (Fig. 4) consists of an input matching network providing the required return loss with minimum insertion

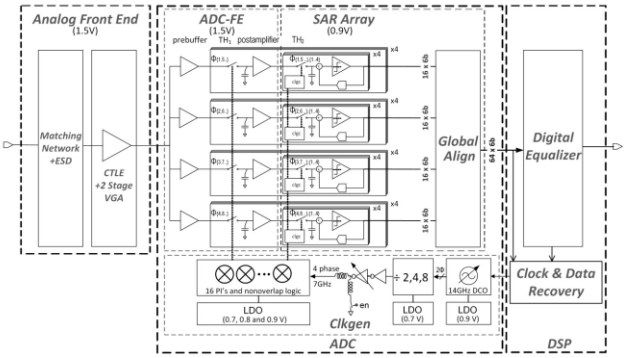



**FIGURE 3.** Overall receiver architecture.

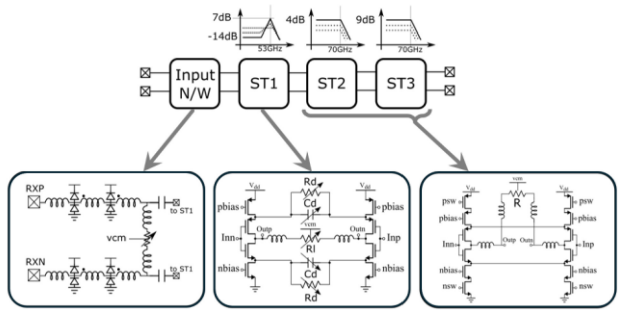



**FIGURE 4.** AFE circuit.

loss up to Nyquist frequency, a hybrid $Q$-shaping-based CTLE which performs most of the coarse ISI cancelation and two stages of programmable wideband the actual SNR benefit VGAs providing most of the AFE gain.

The CTLE's hybrid architecture combines inductive peaking and conventional RC degeneration to obtain both high and mid-band channel equalization. The complementary transconductance (Gm) topology provides better power efficiency compared with a conventional nMOS-based topology using a 1.2-V supply. The VGAs have a topology similar to that of the CTLE but without source degeneration or $Q$-shaping inductive load. Instead, the stage has a series-shunt inductive peaking load which is meant to extend the bandwidth of the VGA. The final VGA stage drives four prebuffers in the ADC front-end.

Each prebuffer in the ADC front-end drives four TH1 switches. To minimize the prebuffer's input capacitance and maximize its bandwidth, the prebuffer's gain is chosen to be 0 dB. At such high-sampling speeds, linearity benefits from boot-strapped track-and-hold switch topologies are minimal at best, with a high power and signal loading costs. Therefore, a simple pMOS switch was used to implement the TH1 switch as shown in Fig. 5(a). The sizing and layout of the switch were picked to minimize signal feedthrough and clocking power. Like the VGA stages, inductive peaking was used at the prebuffer to improve its bandwidth and compensate for the bandwidth limitations of the TH1 switch. Aperture jitter and sampling distortions are minimized using clocks with fast slopes and high swings.

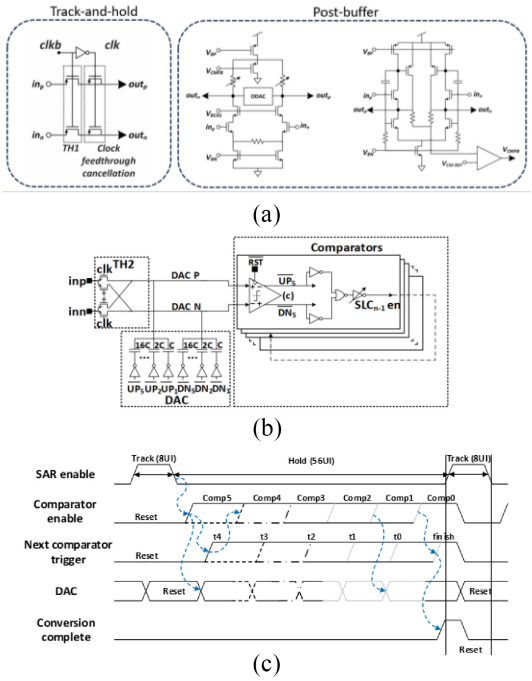



**FIGURE 5.** ADC components, including (a) sampling switch (track-and hold) and post-buffer, (b) loop unrolling SAR-ADC, and (c) its corresponding timing diagram.

The clocks driving TH1 are generated by 16 PIs, ensuring the 1UI spacing between the adjacent clock phases. Each PI is driven by 2 or 3 in-quadrature 7-GHz clock phases, generated using a quadrature divider from a 14-GHz DCO whose frequency and phase are controlled by the CDR. To drive the large capacitive load of the 16 PIs and to provide sinusoidal clocks at the inputs of each PI with minimal random jitter (RJ), a clock driver with a shunt-series inductive peaking topology is used.

Following each TH1 is a post-buffer which drives four SAR-ADC units [9]. The post-buffer consists of two stages as shown in Fig. 5(a): 1) a gain stage and 2) a driving stage. The gain stage provides up to 3-dB gain and consists of an nMOS-based differential pair followed by an nMOS cascode and a differential tunable resistive load. The cascode topology provides better reverse isolation, reduces the Miller effect, and enhances the output bandwidth. The load is tuned to obtain the required gain of 3 dB. The driving stage consists of a source follower with multiple feed-forward loops to extend its bandwidth and enhance its large signal driving capabilities for rising and falling output voltages.

The SAR-ADC uses a loop unrolled (LU) topology to meet the high-speed requirements for a single SAR of 1.75GS/s as shown in Fig. 5(b) [4]. In a LU SAR the number of comparators matches the number of SAR bits. The data gets stored at the comparator itself, eliminating the need for memory register units between comparator and DAC. This also eliminates the need for comparator reset after each conversion as in indicated by Fig. 5(c). As a result, a faster SAR conversion speed can be achieved being limited only by the comparator's clock to output delays and DAC settling

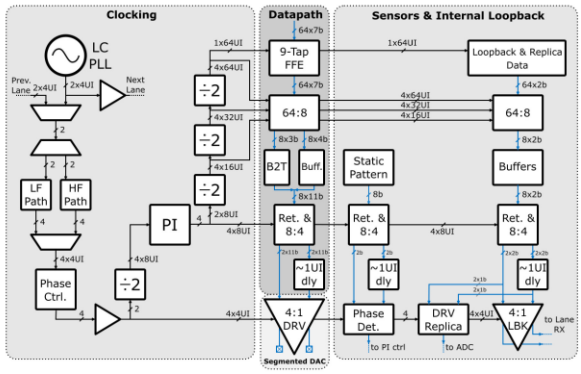



**FIGURE 6.** Overall transmitter architecture.

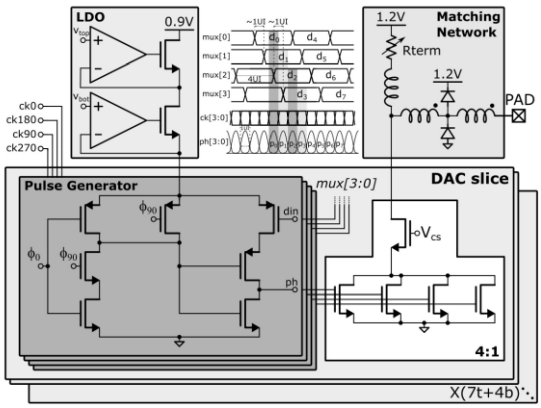



**FIGURE 7.** Core TX driver circuitry.

time. Under the 7-bit mode configuration, the SAR-ADC can achieve an ENOB of up to 5.84 at Nyquist [10].

## IV. 224G TRANSMITTER

The 224G TX consists of three main building blocks: 1) data-path; 2) clock-path; and 3) auxiliary circuits and loop-back path as shown in Fig. 6 [3], [11]. The data-path consists of multiple stages of serialization and digital signal conditioning until the final driver stage where a final 4:1 serialization is performed. The digital data is first processed by a digital FFE with programmable partition between precursors and post-cursors. The 7-bit 64-symbol binary output of the digital FFE is fed to a 64:8 serializer and 3 of the serializer's bits are further converted to thermometric bits. The binary and thermometric outputs of the serializer are then sampled by an 8UI retimer and passed to another 8:4 serializer. To avoid another power intensive retiming stage, a 1UI delay element is used to properly align the data for the final serializer stage. The element is programmable to guarantee adequate delay across PVT variations.

The clock-path consists of LC-DCO-based PLL, quadrature clock generation, clock distribution and buffering, and multiple stages of dividers to generate the required clock speeds for the various stages of serialization. The LC-DCO with a 156.25-MHz reference is based on a complementary transconductance structure with supply and ground LC filters for supply noise rejection and improved phase noise

performance. An inverter-based delay line is used to generate quadrature clocks for 112Gbaud sampling speed. The duty-cycle of the quadrature clocks are calibrated using the phase control block. Per-phase series-shunt peaking inductors at the end of the clock distribution provide both band-pass jitter filtering and bandwidth extension, thus reducing clock distribution power consumption by allowing the use of higher clock drivers fan-out. A PI is used to guarantee the best timing margins on the interface between the 8:4 and the 4:1 MUX/driver.

The auxiliary circuits are used for clock calibrations. During the clock calibrations, different clock patterns are sent to a replica output driver which feeds a low-pass filter followed by a high-resolution, low-noise (<80uVrms) ADC, allowing an accurate correction of any misalignment between the four phases to minimize even-odd jitter (EOJ). After calibrations are completed, the driver starts transmitting while the replica driver becomes responsible in generating the above patterns to keep track of phase misalignment due to temperature drift. A phase detector (PD) on the replica path is used to control the PI. And a replica data-path is used to provide the signals for the internal TX–RX loopback channel.

To reduce the power consumption and jitter/ISI amplification, the core TX driver block, shown in Fig. 7, does not include a predriver stage and instead places the final 4:1 serializer directly at the output stage. The pseudo-differential driver includes 4:1 MUX based on grounded switches with series weighted current sources sized for optimal output impedance with the smallest drain capacitance. They are all biased by the gate voltage Vcs, calibrated to reach the target 1-Vpp differential swing on a 100-Ohm termination. The series switches are driven by the 1UI pulse generator that combines the relevant data with a quadrature-generated 1 UI pulse. The output pad is connected to an inductive matching network, ESD protection diodes, and tunable termination resistor with peaking inductor for bandwidth extension.

## V. SIGNAL INTEGRITY CONSIDERATIONS—DEMANDING CHANNEL, INCLUDING OSFP CONNECTORS AND 1-M CABLE

Real-world implementation of end-to-end (E2E) links defined by 200G IEEE Ethernet and OIF compliance standards is challenging. A typical link is composed of multiple components: package, PCB, cables, and connectors. Each component poses its unique implementation and modeling challenges. Therefore, accurate analysis methods predicting the link performance are essential to achieve good correlation between prediction and lab measurements. In this section, analysis and design optimization for performance enhancement on E2E channel component are discussed.

As illustrated in Fig. 8, the channel consists of a package, socket, PCB, OSFP connector, and a 1-m passive copper cable assembly. Each component has been designed and optimized to meet the bandwidth, return loss, and crosstalk

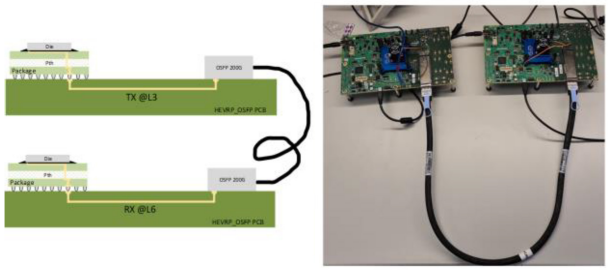



FIGURE 8. Simulated channel configuration.

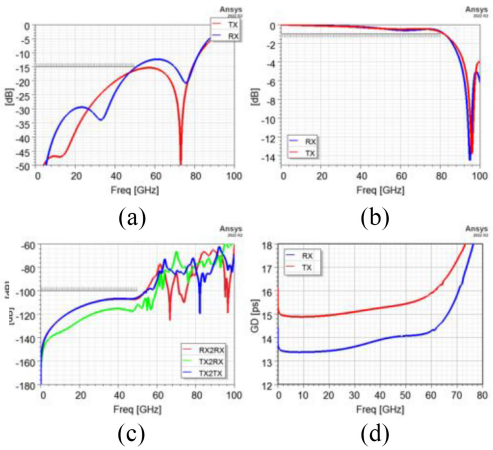



FIGURE 9. Simulated package performance: (a) Differential return loss, (b) differential insertion loss, (c) crosstalk, and (b) group delay.

requirements necessary for successful transmission. The 224-Gb/s PAM4 LR channel is designed to achieve up to 40-dB insertion loss at 56 GHz from bump to bump.

### A. 224-GB/S PACKAGE DESIGN OPTIMIZATION

As discussed in [12], several technology advancements need to happen to scale the package with the same or similar loss at Nyquist of 224-Gb/s PAM-4, including better new material, improved surface roughness, finer BGA pitch, use of skip-layer routing design technique, better trace line/space, better core thickness, and stack-up. Based on the studies, 0.8-mm BGA pitch can be used to support high-enough insertion loss bandwidth and avoid unwanted resonances caused by higher order mode propagation.

Our 224-Gb/s package design used a 12-layer organic stack-up, and the 0.8-mm pitch BGA was optimized to meet better than 15-dB return loss up to 50 GHz. Simulated differential return loss, insertion loss, cross talk, and group delay of the package trace-via-PTH-BGA ball transition are presented in Fig. 9. The trace to BGA transition achieves 1-dB insertion loss and 15-dB return loss and less than −100-dB cross talk up to 50 GHz and less than −70 dB up 60 GHz. The group delay variation is less than 1 ps up to 60 GHz. The effective ball/pad capacitance is smaller than 30 fF. To achieve package loss targets, a skip-layer trace configuration is required in the global routing area to attain an average trace loss below 0.15 dB/mm at 56 GHz.

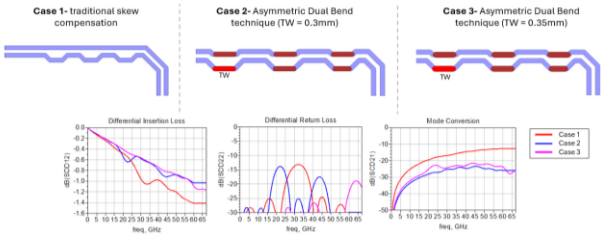



**FIGURE 10.** Asymmetric dual bend comparison.

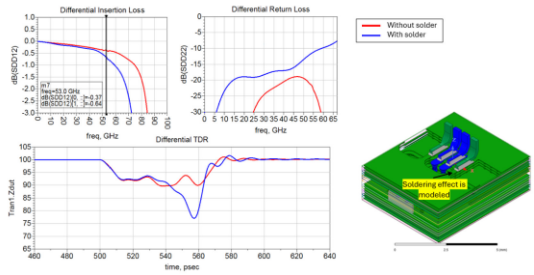



**FIGURE 11.** Connector performance with and without solder.

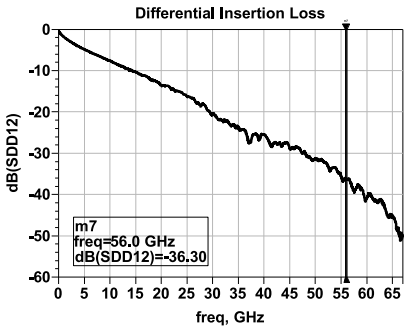



**FIGURE 12.** Simulated full channel's differential insertion loss.

### B. 224-GB/S PAM4 PCB DESIGN

The PCB design is a critical component of the 224-Gb/s PAM4 channel. To minimize trace loss, our PCB incorporates a 0.8-mm ball pitch, a skip-layer design, and high-quality materials, such as smooth copper (HVLP type) and Isola Tachyon, an ultralow loss dielectric material. This combination enables a low-insertion loss of 1.5 dB/inch at 56 GHz [13].

Traditional skew compensation techniques involving bends often introduce imbalances between the differential $P$ and $N$ traces due to asymmetrical 45° corners. These corners create additional parasitic elements in one trace, leading to differential-to-common mode conversion. To mitigate this issue, we adopted the asymmetric-dual-bend technique [14]. As shown in Fig. 10, this approach significantly improves SCD21 by approximately 10 dB for frequencies up to 60 GHz in strip-line routing.

### C. OSFP CONNECTOR AND CABLE ASSEMBLY

A comprehensive evaluation of various cable and connector configurations were conducted. Due to backward compatibility requirements with 112-Gb/s standards, the projected loss target for a 1-m cable assembly, including two connectors, is estimated to be around 15 dB based on current technology limitations. While a detailed analysis of cable assembly design is beyond the scope of this article, it is evident that technological advancements and stringent manufacturing controls are necessary to achieve an improved loss target of approximately 10 dB at the Nyquist frequency for a 1-m cable with two connectors.

Connector stub tolerance and soldering processes significantly impact overall performance. As highlighted in [15], these factors must be carefully considered to prevent excessive insertion loss variation and return loss degradation. Fig. 11 illustrates the critical impact of soldering on connector performance, demonstrating a potential impedance drop of 10 ohms and more than 5-dB degradation in differential return loss performance when solder is not accounted for in the connector optimization process.

### D. E2E CHANNEL CORRELATION

Accurate characterization of real-world channels is a complex task. To achieve reliable measurement results, precise vector network analyzer (VNA) calibration utilizing small 100-$\mu$m pitch pico-probes is essential. To correlate simulation models with measured data, an accurate co-modeling methodology for the package and PCB is used. This approach effectively avoids double counting of BGA balls and accurately represents the vertical transitions between the package and PCB. Similar techniques are applied to the connector interface. As depicted in Fig. 12, the channel is simulated to achieved 36-dB insertion loss at 56 GHz from bump to bump.

## VI. ADVANCED APPLICATIONS: DIRECT DRIVE LINEAR OPTICS

While the SerDes was initially designed to operate over an electrical channel, its advanced DSP capabilities allow it to function efficiently even when an optical link is incorporated between the TX and RX. The feasibility of this design is attributed to the stringent linearity and bandwidth requirements of the optical system.

To extend the operation of the SerDes into the optical domain, the electro-optics setup depicted in Fig. 13 was constructed. As can be seen in the figure, the electrical SerDes transceiver embedded within a system on chip (SOC) interfaces with external electrical integrated circuit components (EIC), which in turn interface with the photonic integrated circuits (PICs) which together with a 23-km fiber optical cable make up the optical link.

The transmit optical link consists of the distributed-feedback laser (DFB), Mach–Zehnder modulator (MZM), optical equalizer, 23 km of standard single mode fiber and photodiode. A balun is used to convert the TX's output from a differential to a single-ended signal which gets buffered using an off-chip ultrabroadband amplifier which in turn drives the MZM. The O-band DFB emits light at a wavelength of 1310 nm as an input source

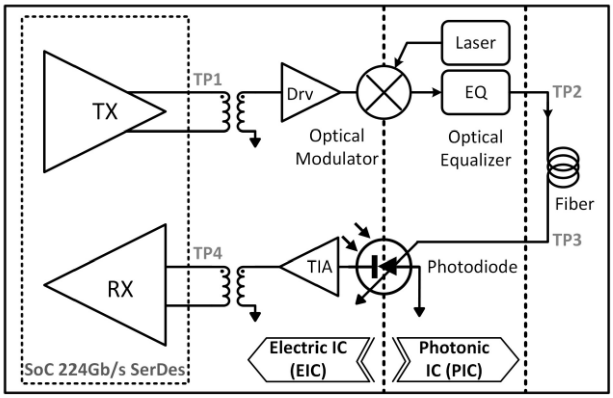



**FIGURE 13.** Direct drive linear optics SerDes implementation.

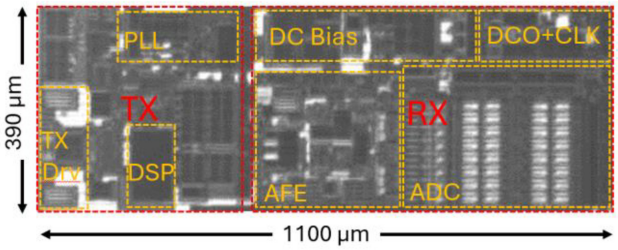



**FIGURE 14.** Die photograph of transceiver.

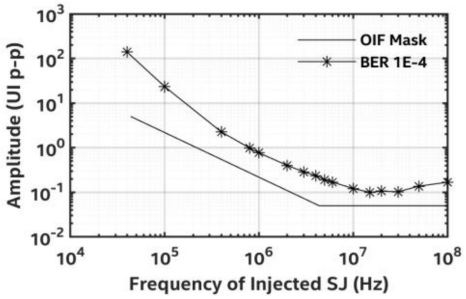



**FIGURE 15.** Jitter tolerance of the receiver under 40-dB insertion loss channel at 56 GHz.

light to the MZN. The relative intensity noise (RIN) of the DFB is about −145 dB/Hz. The operating current of the laser is 100 mA, with a corresponding emitted power of 35 mW.

A practical MZM is designed with a limited BW of about 30 GHz. The optical equalizer plays a crucial role in mitigating both baseband and optical impairments, such as the limited MZM BW. In this scenario, the channel bandwidth is predominantly constrained by the cutoff frequency of the modulator. Despite the chromatic dispersion being nearly negligible at a wavelength of 1310 nm, the high bandwidth of the RF signal still necessitates some degree of dispersion compensation for the 23 km of fiber.

The transmitted optical signal is detected by the photodiode (PD). The PD is designed to achieve high bandwidth of > 60 GHz with sharp roll-off for noise reduction. Finally, the PD's induced current is fed to the TIA which drives the Serdes.

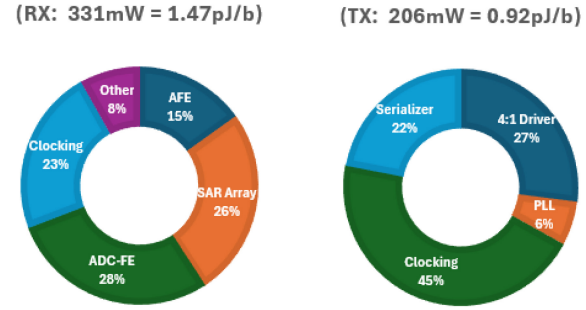



**FIGURE 16.** Breakdown of the 537-mW total transceiver analog power (2.39 pJ/b) between the RX and TX components.

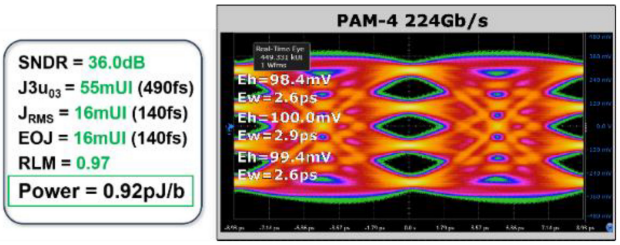



**FIGURE 17.** Transmitter's PAM4 eye-diagram and compliance merits.

## VII. MEASUREMENTS AND RESULTS

The transceiver was fabricated in a 3-nm CMOS process and the die photograph of it is shown in Fig. 14. The full transceiver measures approximately 1100 $\mu$m × 390 $\mu$m. The TX measures about 390 $\mu$m × 390 $\mu$m, and the RX occupies the rest. The major TX and RX blocks are also marked on the die photograph, including the TX driver, PLL, DSP block of the TX, AFE, ADC, and DCO, including dividers and clock distribution drivers and DC bias circuits.

### A. WIRELINE TRANSCEIVER MEASUREMENTS

#### 1) 224-GB/S RX MEASUREMENTS

The receiver was characterized using a BER tester (BERT), Keysight M8050A, with a 224-Gb/s PRBS pattern. A BER of 2.5e-8 was achieved on a nonreflective channel with a bump-to-bump insertion loss of 40 dB at 56 GHz. The BERT's swing was set to a 1.2-V diff p-p. The jitter tolerance measurement demonstrates a CDR loop with a 5-MHz bandwidth and an eye opening of 0.1UI for 1e-4 BER as shown in Fig. 15. This RX achieves 1.47-pJ/b energy efficiency at 224 Gb/s excluding DSP power as shown in Fig. 16. The performance of the RX versus similar 112/224-Gb/s RX's is summarized in Table 1.

To test the advantage of improved ADC resolution, a similar BERT PRBS test was conducted on a medium-reach channel (20-dB insertion loss at Nyquist). The ADC resolution was toggled between 6 and 7-bit modes. The 6-bit BER of 5.1e-10 had an insignificant drop to 4.0e-10 under this condition. For medium and long-reach channels, the ADC's resolution has minor impact on BER since the RX's performance is dominated by the AFE's noise and linearity rather than the ADC's resolution and quantization noise.

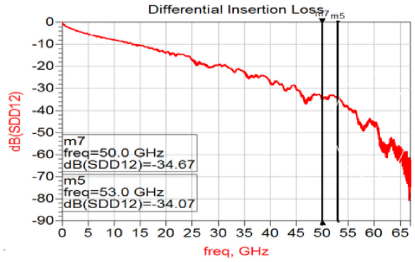



**FIGURE 18.** Measured insertion loss of OSFP connector 1-m cable channel.

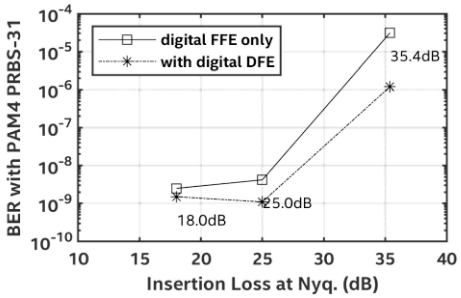



**FIGURE 19.** BER versus insertion loss at 56 GHz with digital FFE enabled and with digital DFE enabled.

### 2) 224-GB/S TRANSMITTER MEASUREMENTS

The TX performance was evaluated with an on-package-connector (OPC) connected through a 6-inch coaxial cable and DC blocker to a 256GS/s Keysight UXR1104A real-time oscilloscope. The package loss was de-embedded by the measured package *s*-parameters and no additional scope equalization was employed.

Fig. 17 shows eye diagrams, compliance parameters and energy efficiency of transmitted data in PAM-4 224 Gb/s, with on-chip FFE set to maximize eye opening. The eye width (Ew) and eye height (Eh) are reported in the figure for a target BER of 1e-4, showing a central eye vertical opening of 100.0 mV at 224-Gb/s PAM-4. This TX achieves 0.92-pJ/b energy efficiency at 224 Gb/s excluding DSP power as shown in Fig. 16. The performance of the TX versus similar 224-Gb/s TX's is summarized in Table 1.

### 3) TRANSCEIVER MEASUREMENTS: OSFP REFLECTIVE CHANNEL AND DFE VERSUS FFE ONLY

The transceiver was tested with the reflective OSFP assembly, consisting of two OSFP connectors and a 1-m cable as depicted in Fig. 8. The channel's insertion loss was measured to be 34 dB at 50 GHz (Fig. 18). At a PAM4 200 Gb/s, the system managed to achieve a BER of 1e-5 under this configuration.

To characterize the impact of DFE plus FFE versus FFE only equalization, a 224-Gb/s transceiver measurement was conducted with an external loop with channel configurations corresponding to 18, 25, and 35.4-dB insertion losses at 56 GHz. The loopback BER versus insertion loss is depicted in Fig. 19. It can be seen from the figure that enabling the single tap digital DFE can give up to an order of magnitude

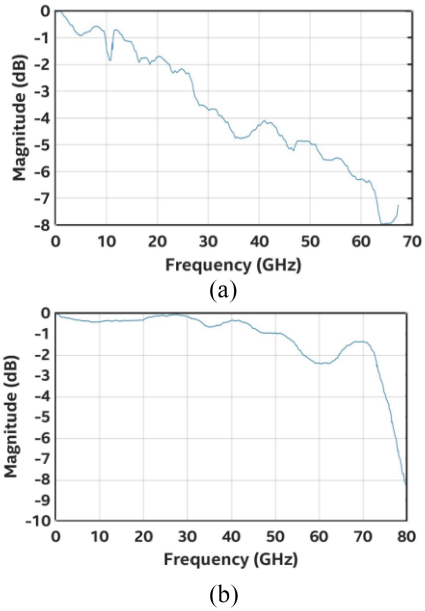



**FIGURE 20.** Characterization of optical link components showing (a) MZM's electro-optical transfer function and (b) photodiode's frequency response.

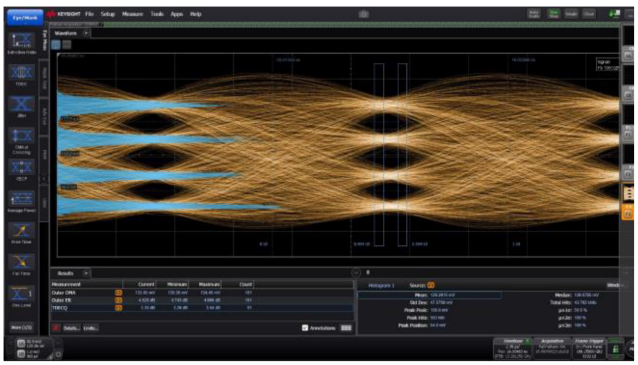


**FIGURE 21.** Optical eye diagram of the optical link with a 23-km length of fiber.

of BER enhancement for longer reach channels. It is less effective for shorter reach channels as the residual ISI post FFE is smaller and the BER is dominated by receiver noises rather than residual ISI.

## *B. DIRECT DRIVE LINEAR OPTICS MEASUREMENTS*

To evaluate the electro-optical system, its components were measured separately. The MZM modulates the DFB's signal with an electro-optical transfer function depicted in Fig. 20(a). The 3 and 6-dB electro-optical cut-off frequency is about 30 and 60 GHz, respectively. The MZM is designed with a near-zero linewidth enhancement factor ($\alpha$), resulting in minimal induced chirp. This dramatically reduces the impact of chromatic dispersion on system performance, enabling longer distance signal transmission. The optical signal, once transmitted, is detected by the photodiode (PD). The custom-designed and fabricated PD achieves a 3-dB bandwidth of approximately 70 GHz, as shown in Fig. 20(b), and a responsivity of 0.65 A/W within the O-band. Notably,

**TABLE 1.** RX and TX summary and comparison.

| | | JSSC'22 [16] | JSSC'20 [17] | JSSC'24 [4] | ISSCC'24 [18] | ISSCC'24 [19] | This work |
|---|---|---|---|---|---|---|---|
| | **Data rate** | 224Gb/s | 112Gb/s | 224Gb/s | 224Gb/s | 212Gb/s | 224Gb/s |
| | **Technology** | 10nm | 10nm | 5nm | 3nm | 5nm | 3nm |
| | **Power Supply (V)** | 0.85,1,1.5 | 0.9,1.5 | 0.9,1.5 | 0.75,1.1,1.2 | 0.65,0.94,1.2 | 0.75,0.9, 1.2 |
| **RX** | **Power* (pJ/b)** | | 4.2 | 1.41 | 2.0 | 1.36 | 1.47 |
| | **Ch. loss @ Nyq. (dB)** | | 35 | 38 | 40 | -- | 40 |
| | **BER** | | 1.00E-06 | 6E-7 | 1E-6 | 1E-12 | 2.5e-8 |
| | **Area ($mm^2$)** | | 0.281 | 0.34 | 0.22 | 0.72 | 0.277 |
| **TX** | **Power* (pJ/b)** | 1.88 | | | 1.0 (w/o PLL) | 1.33 | 0.92 |
| | **Swing (Vpk-pk)** | 1.0 | | | 0.96 | 0.78 | 1.0 |
| | **RJ ($fs_{rms}$) @Nyq** | 65 | | | 55 | 72 | 62 |
| | **SNDR (dB)** | 33.3 | | | | -- | 36 |
| | **RLM** | 0.99 | | | 0.99 | 0.98 | 0.97 |
| | **J3u$_{03}$ (mUI)** | | | | | | 55 |
| | **Area ($mm^2$)** | 0.18 | | | 0.1 (w/o PLL) | 0.64 | 0.15 |

* Does not include the DSP power




**FIGURE 22.** Loopback histogram with a BER of 1e-4 for a direct drive optical channel with 23-km fiber cable.

the PD features a highly linear response, which helps to preserve signal integrity throughout the detection process. Additionally, the sharp roll-off of the PD's response curve is beneficial in reducing noise levels. Finally, the PD's induced current is fed to the TIA which drives the RX through a single-to-differential balun.

The electro-optical eye diagrams measured between TP1 and TP4 (Fig. 13) with a 23-km optical fiber is shown in Fig. 21. The measured TX dispersion eye closure quaternaries (TDECQs) are 3.3 and 3.48 dB; the extinction ratios are 5 and 5.9 dB; all four levels are very well separated and distinguished. The power consumption of the optical system is about 2.25 pJ/bit. The optical link was combined with the SerDes in a TX to RX external loopback fashion. The corresponding histogram in Fig. 21 shows the well-separated four levels of PAM4 signal and related BER is in order of 1e-4 (Fig. 22).

## VIII. CONCLUSION

This article presented the system considerations, circuit architecture, and design implementation of a beyond 200-Gb/s wireline and linear optics transceivers. A transceiver capable of supporting long-reach channels (>35 dB at Nyquist) was fabricated in an advanced 3-nm CMOS process. The results demonstrate the technology's benefits of doubling the data rate of transceivers while achieving efficiency gains in power consumption and silicon area. The proof-of-concept demonstration of 224 Gb/s with linear optics opens the avenue for power-efficient, low-latency future optical communication.

## ACKNOWLEDGMENT

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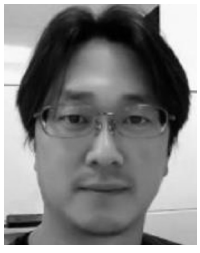
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